ECG Signal Denoising Using Digital Filtering and Wavelet-Based Methods

ELEC5305 Project Feedback Two

Updated progress report based on Feedback 1 proposal

1. Student Information

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GitHub Project Link: https://github.com/shayujie2002-wq/elec5305-project-540605336
GitHub Pages Link: https://shayujie2002-wq.github.io/elec5305-project-540605336/

2. Project Overview

This project investigates ECG signal denoising using conventional digital filtering and wavelet-based methods. The aim is to reduce common interference such as baseline wander, 50 Hz power-line interference, and high-frequency random noise while preserving important ECG waveform features, particularly the QRS complexes.

Since Feedback 1, the proposed processing pipeline has been implemented in Python and tested on ECG data from the MIT-BIH Arrhythmia Database using PhysioNet/WFDB. Controlled noise is added to a selected ECG segment so that denoising performance can be evaluated against a known reference. Conventional filtering and a preliminary discrete wavelet transform (DWT) method are now compared quantitatively using signal-to-noise ratio (SNR), mean squared error (MSE), and root mean squared error (RMSE).

3. Background and Motivation

ECG recordings are sensitive to several forms of interference. Low-frequency baseline drift can shift the waveform over time, power-line interference can introduce a narrow-band component around 50 or 60 Hz, and muscle or electronic activity can contribute broadband high-frequency noise. These disturbances may reduce the reliability of later feature extraction, heart-rate estimation, and automated ECG analysis.

Digital filtering provides a computationally efficient method for suppressing noise components that occupy known frequency ranges. A notch filter can target power-line interference, while a band-pass filter can reduce both baseline drift and high-frequency noise. Wavelet denoising is also relevant because ECG is a non-stationary biomedical signal and wavelet analysis represents signal content at multiple time and frequency scales. The project therefore compares these two approaches rather than assuming that one method is universally superior.

4. Current Methodology

The implementation is written in Python using NumPy, SciPy, Matplotlib, Pandas, PyWavelets, and WFDB. MIT-BIH record 100 is loaded through PhysioNet, and a 10-second segment beginning at 5 seconds is used for the current preliminary experiment.

- Noise model: 0.33 Hz baseline wander, 50 Hz sinusoidal power-line interference, and additive random high-frequency noise.
- Conventional filtering: a 50 Hz IIR notch filter (Q = 30), followed by a fourth-order 0.5-40 Hz Butterworth band-pass filter.
- Wavelet denoising: DWT using the db6 wavelet, decomposition level 5, and soft thresholding based on an estimated noise level.

- Evaluation: SNR, SNR improvement, MSE, and RMSE, together with time-domain, frequency-domain, and morphology comparisons.

5. Preliminary Outcomes

The current experiment produced clear differences between the two denoising approaches. The noisy ECG has an SNR of -2.13 dB. Conventional digital filtering increases the SNR to 11.18 dB, giving an improvement of 13.31 dB, while reducing RMSE from 0.17093 to 0.03694. The current wavelet configuration improves the SNR by only 1.22 dB and reduces RMSE to 0.14857.

Method	SNR (dB)	MSE	RMSE	SNR Improvement (dB)
Noisy ECG	-2.13	0.02922	0.17093	0.00
Digital filtering	11.18	0.001365	0.03694	13.31
Wavelet denoising	-0.91	0.02207	0.14857	1.22

Table 1. Preliminary denoising performance on the current MIT-BIH ECG segment.

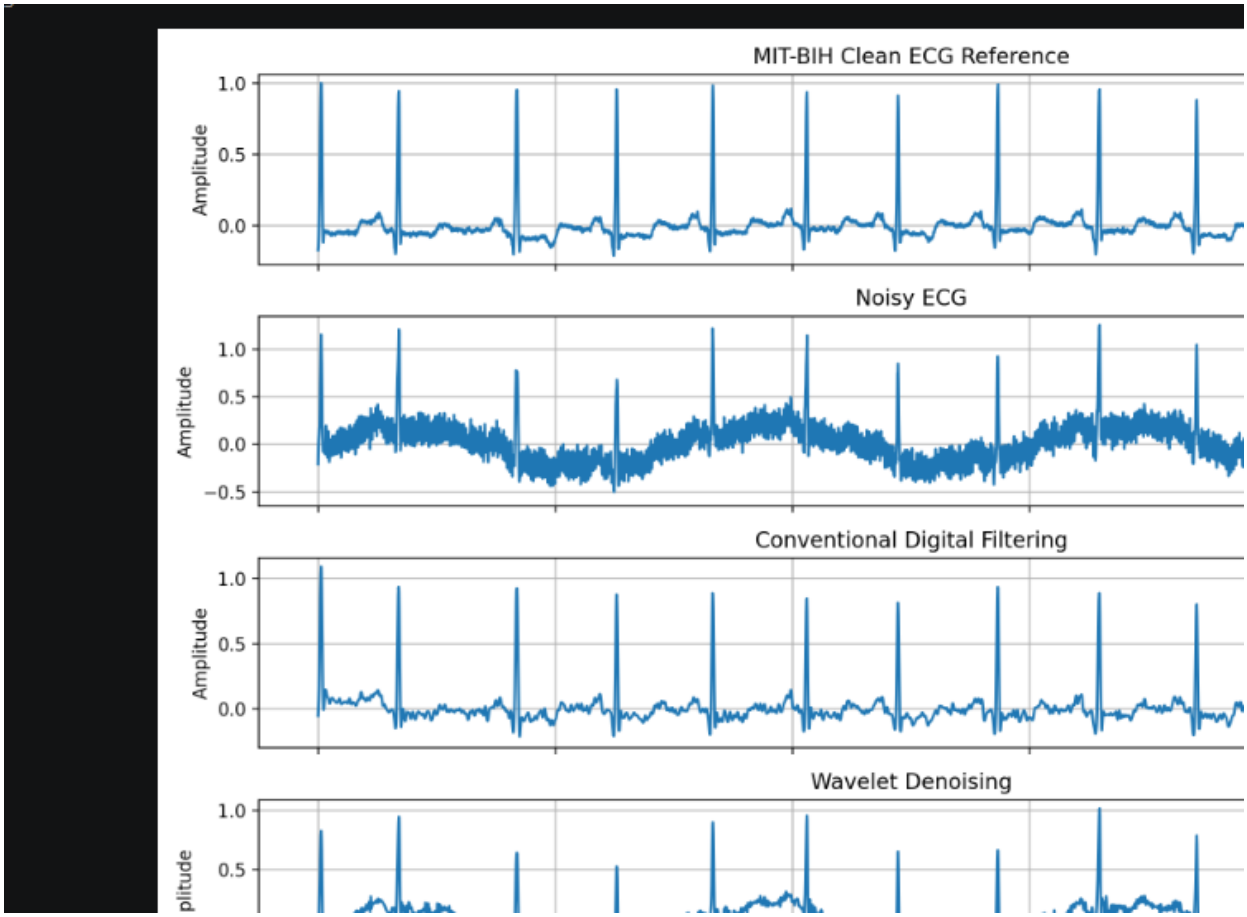


Figure 1. Time-domain comparison of clean, noisy, digitally filtered, and wavelet-denoised ECG signals.

The time-domain result shows that the conventional digital filter removes most of the baseline drift and preserves the major ECG peaks closely. The current wavelet result removes some noise but retains noticeable low-frequency drift and alters some waveform amplitudes. The frequency-domain comparison also shows strong suppression of the 50 Hz interference by conventional filtering.

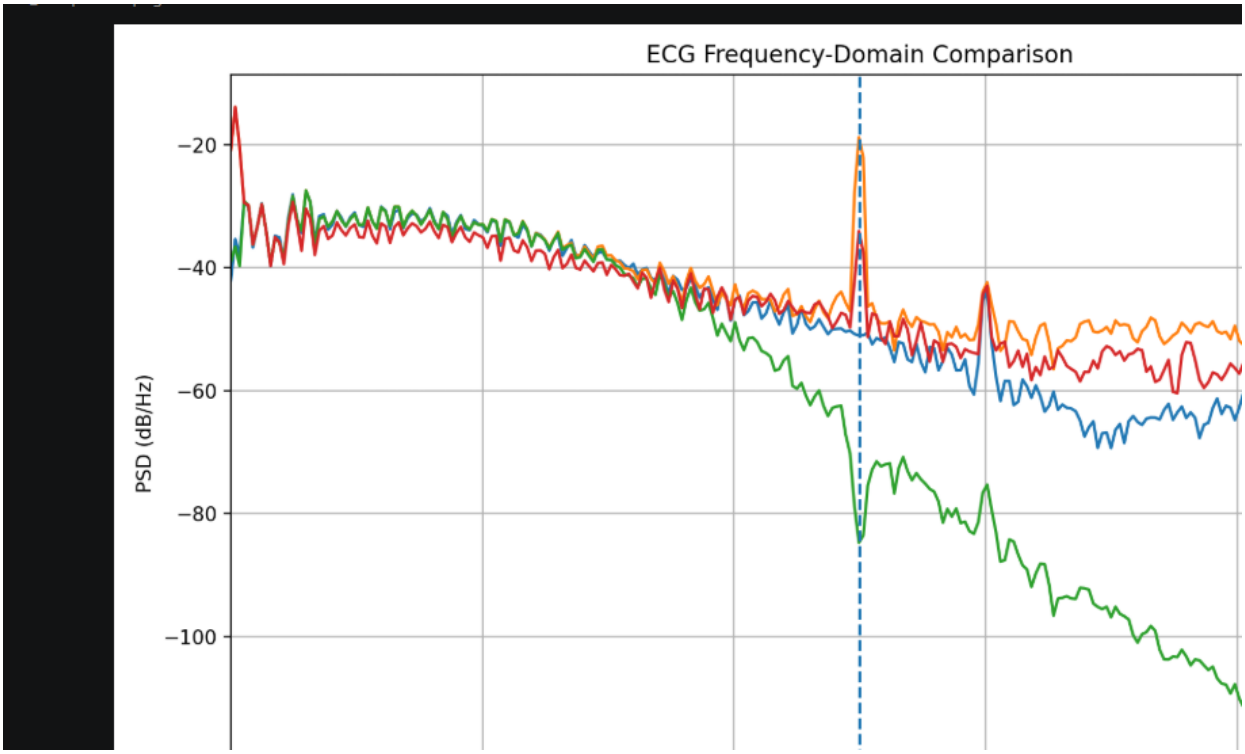


Figure 2. Power spectral density comparison, including the 50 Hz interference component.

6. Updated Timeline

Week	Status	Task / Progress
Weeks 1-2	Completed	Finalised project scope, created the GitHub repository, and selected ECG denoising as the project topic.
Weeks 3-5	Completed	Reviewed ECG noise sources, identified MIT-BIH/PhysioNet as the data source, and selected SNR, MSE, and RMSE as evaluation metrics.
Weeks 6-7	Completed	Implemented ECG loading, visualisation, controlled noise generation, and basic digital filtering in Python.
Weeks 8-9	In progress / current	Implemented 50 Hz notch filtering, 0.5-40 Hz Butterworth filtering, initial DWT wavelet denoising, and preliminary quantitative comparison.
Weeks 10-11	Next	Optimise filter cutoff frequencies and orders; compare wavelet families, decomposition levels, and threshold strategies; test additional MIT-BIH records.
Weeks 12-13	Planned	Complete final evaluation, analyse waveform preservation and computational performance, update GitHub documentation, and prepare the final report.

7. References

1. Thakor, N. V., & Zhu, Y. S. (1991). Applications of adaptive filtering to ECG analysis: Noise cancellation and arrhythmia detection. IEEE Transactions on Biomedical Engineering, 38(8), 785-794.
2. Addison, P. S. (2005). Wavelet transforms and the ECG: A review. Physiological Measurement, 26(5), R155-R199.
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4. Goldberger, A. L., et al. (2000). PhysioBank, PhysioToolkit, and PhysioNet: Components of a new research resource for complex physiologic signals. Circulation, 101(23), e215-e220.